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APPLIED RESEARCH

Real-World Gaps in AI Governance Research

ILAN STRAUSS[1,2,3], ISOBEL MOURE[1], TIM O'REILLY[1,4], AND SRULY ROSENBLAT[1]

[1]AI Disclosures Project, Code for Science and Society, New York, NY 10004, USA
[2]Institute for Innovation and Public Purpose, University College London, WC1B 5BP London, U.K.
[3]SARChI-ID, University of Johannesburg, Johannesburg 2006, South Africa
[4]O'Reilly Media, Sebastopol, CA 95472, USA

Corresponding author: Ilan Strauss (ilan@aidisclosures.org)

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ABSTRACT Drawing on 1,178 AI risk and reliability papers from 9,439 generative AI papers (January 2020 through March 2025), we compare research outputs from the most influential research institutions: frontier *AI companies* (Anthropic, Google DeepMind, Meta, Microsoft, and OpenAI) and leading *AI universities* (CMU, MIT, NYU, Stanford, UC Berkeley, and University of Washington). We find that Frontier Corporate AI research increasingly concentrates on pre-deployment areas — model alignment and testing & evaluation — while attention to deployment-stage issues, such as model bias, has waned as commercial imperatives and existential risk concerns have taken precedence. We identify significant research gaps in high-risk deployment domains, including healthcare applications, commercial and financial contexts, misinformation, persuasive and addictive features, hallucinations, and copyright usage in training and inference. Without concerted efforts to enhance external observability into AI's deployment, the growing concentration of AI research with frontier corporations could deepen knowledge deficits in these critical deployment areas. We recommend measures to expand external researcher access to deployment data and improve systematic observability of AI systems' in-market behaviors.

INDEX TERMS AI governance, AI safety, reliability, generative AI, deployment, policy.

I. INTRODUCTION

As generative AI becomes integrated into every facet of our work and social lives, risks are emerging in everyday commercial ''post-deployment'' contexts [1], [2]. Torts and litigation are rising, teens have committed suicide, and hallucinations are causing reliability problems [3]. But we have little hard data on who or what is responsible for these effects.

This observability gap, we show, is due to what frontier corporate AI safety & reliability research chooses to focus on and expose. A handful of frontier AI corporations increasingly defines the agenda for public AI risk investigations [4] and focuses on model alignment and testing in *pre-deployment* laboratory settings [5] – Figure 4.[1] User, system, and society-level impacts remain neglected.

Outside of China, the large frontier models, the Ph.D. graduates, the incredible stores of compute, the LLM benchmarks, and the R&D resources increasingly reside with a handful of U.S. corporations. Industry's share of the biggest, and therefore the most capable, AI models went from 11% in 2010 to 96% in 2021 [4]. In 2020, nearly 70% of AI Ph.D. holders from North American universities were recruited by industry, up from about 21% in 2004 [6]. In 2023, the U.S. higher education sector spent $102 billion on R&D [7], but this was still dwarfed by Big Tech which exceeded $170 billion.[2] Google in particular is fourth in the world among all companies for research outputs in the OpenAlex database (see also [9]).

However, real-world risk areas, including those currently considered highest risk by AI companies themselves, are

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[1]AI alignment covers 'post-training' interventions, fine-tuning & reinforcement learning from human and AI feedback.

[2]Based on 10-K and 10-Q reports: Apple (FY 2023, year ended Sept 30, 2023): $29.9 bn; Alphabet (2023): $45.4 bn. Microsoft (FY 2024, year ended June 30, 2024): $29.5 bn in 2024, $27.2 bn in 2023; Meta Platforms (full-year 2023, from 2023 results / 10-K): $38.5 bn in 2023. Amazon does not break out R&D cleanly but estimates put it above $45 bn [8].

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often neglected in frontier lab research [10]. These include risks from model persuasiveness, emergent behaviors from reinforcement learning exploitation ('reward hacking'), and misinformation [11], [12], [13]. Frontier lab alignment research has largely overlooked dangerous side-effects, such as sycophancy and degraded answer quality [14], [15], [16], [17].

A. MOTIVATION

The research presented in this paper is motivated by three observations:

(1) There is a growing disconnect between the theoretical research being prioritized at the major corporate AI labs, which examines AI models in isolation, and the growing need for research on how AI systems function in real-world deployment contexts where their outputs vary greatly by prompt, context, and implementation [18], [19], [20].

(2) Commercial activity is itself a major source of risk in post-deployment AI systems, yet those in the best position to monitor and understand those risks have economic and reputational incentives to underplay them, rather than conduct transparent research on emerging problems.

(3) More and more agenda-setting AI research comes from a small group of incredibly well-resourced corporations, with enormous relative R&D spending [21] and large concentrations of Ph.D. graduates [6]. This itself reflects a compute divide [22] and means collaborations with AI labs become increasingly important for universities looking to access frontier models and significant compute [22]. Most public researchers have limited access to the data needed to assess risks during real-world deployment.

These observations align with broader patterns documented in science and technology studies: when research capacity concentrates in commercial entities, the questions asked – and not asked – increasingly reflect private rather than public interests [23]. The ''AI safety'' agenda itself has been shaped by particular philosophical and institutional traditions [24], with implications for which risks receive sustained attention. Our empirical mapping provides evidence on this dynamic for generative AI.

Without observability and research on AI systems' behavior and risks 'in the wild', we are flying blind. Research into model safety that only examines behavior under controlled laboratory conditions is fundamentally insufficient, as AI models' risks and safeguards in practice often differ significantly from those in theory [25], [26]. Deployment environments dramatically alter model behavior through varied prompting contexts, fine-tuned applications, API access patterns, and infrastructure differences [18], [27]. Moreover, AI systems may be used in fundamentally different ways compared to during testing. Critical post-deployment components affecting safety include orchestration primitives, data-retrieval layers such as RAG, safety and guardrail services, and observability stacks [28].

A structural misalignment between corporate profit incentives and rigorous safety research exacerbates the deployment research gap. Emerging legal cases highlight this pattern: real-world harms emerge from deployed systems, companies respond with minimal changes or counterproductive measures, and research focus remains predominantly theoretical [3], [29]. Character.AI faces lawsuits over addictive bots encouraging self-harm among teenagers, yet Meta subsequently expanded permissions for explicit romantic role-play with AI bots. OpenAI removed impersonation restrictions for Sora, enabling deepfakes [30], while nearly 30 lawsuits target AI developers over copyright infringement [31].

Existing risk disclosures by frontier AI labs focus almost exclusively on malicious use while ignoring commercial uses that may cause equivalent harm [32], [33], [34]. Anthropic exemplifies this misalignment: while announcing interpretability work to find risks in ''model's inner workings'' [35], they simultaneously documented actual malicious uses including recruitment fraud, malware development, and political influence operations [34]. Yet these documented risks have not triggered proportionate research investment into post-deployment safeguards.

B. METHOD

We analyze generative AI risk and reliability research by leading U.S. institutions using a dataset of 1,178 papers from 9,439 generative AI papers written by five frontier AI companies (Anthropic, Google DeepMind, Meta, Microsoft, and OpenAI), and six leading AI research universities (Carnegie Mellon University (CMU), MIT, New York University (NYU), Stanford, UC Berkeley, and University of Washington) between January 2020 and March 2025. We call these two groups 'Corporate AI' and 'Academic AI', respectively, for short, but really they sample the 'bleeding edge' of AI research – highly influential papers that help set the AI governance agenda for governments, society, and other institutions [36], [37].

Our generative AI research dataset is based on OpenAlex, to which we add research papers from Anthropic and OpenAI's websites [38]. We define leading AI risk research as technical and applied safety and reliability research pre- and post-deployment. In conjunction with OpenAI's o3-mini, we determine if papers are ''safety & reliability'' research, and then classify them into one of eight subcategories. We separately look at generative AI research into high-risk domains (based on paper abstracts and titles), covering medical, finance, commercial, & copyright, misinformation, disclosures, behavioral, & accuracy. Our sample is intentionally limited to U.S.-based institutions that set the agenda for frontier AI development; findings characterize these actors' research portfolios and should not be generalized to the broader global AI research ecosystem.

C. CORE FINDINGS

(1) *Among leading AI research institutions in generative AI risks, frontier corporate research is far more influential*. Anthropic, OpenAI, and Google DeepMind each have significantly more citations for their AI safety & reliability work than any of the major U.S. academic institutions we track. Google DeepMind has more citations for its general generative AI research than the top four AI academic institutions combined.

(2) *As frontier AI companies race to commercialize powerful AI systems, their research priorities are increasingly shaped by business incentives rather than by thorough risk assessments*. Most of the governance research by frontier corporate labs focuses on model performance divorced from its applications. For example, ethics & bias research now only receives attention from leading academic institutions.

(3) *Frontier AI companies severely neglect deployment-stage behavioral and business risks*. Only 4% of frontier corporate AI papers (6% leading Academic AI) tackle high-stakes areas like persuasion, misinformation, medical & financial contexts, disclosures, or core business liabilities (IP violations, coding errors, hallucinations) – despite emerging lawsuits showing these risks to already be material.

D. POLICY CONSIDERATIONS

Structured access is needed into deployed AI systems' telemetry data and artifacts in order for third parties to systematically analyze real-world risks and harms. Monitoring and evaluation of LLMs in real-world environments is now essential to quality assurance (QA), as in 'LLMOps' [28]. But the data used for this is the preserve of corporate *practice*, resulting in society losing visibility into AI's ongoing risks and harms. Academic-industry collaborations are therefore important but don't take place on equitable terms [39]. Privacy-preserved external access to data for approved researchers is a more open, accessible option [40]. This could draw on companies' existing *telemetry data* (logs, traces, & business metrics) and LLM model *data artifacts* (e.g., training/fine-tuning datasets). Emerging LLM monitoring frameworks – such as those from LangSmith, Langfuse, OpenTelemetry, & Weights and Biases – make structured & standardized external API access for researchers increasingly feasible. This, however, may expose corporations to liability, such that safe harbors [41], [42] may be needed to support purpose-built external access.

E. LITERATURE AND CONTRIBUTION

This study contributes to three literatures. First, within *bibliometrics of AI research*, we extend work mapping institutional concentration [4], [43] by introducing a pre-/post-deployment distinction and showing how safety research portfolios differ across frontier labs and universities. Second, in *science and technology studies (STS)*, our findings speak to debates on epistemic capture – the tendency for commercially embedded research agendas to crowd out public-interest inquiry [23], [44]; and on how funding structures shape which questions get asked. Third, we contribute to the emerging *AI governance* literature by providing systematic evidence that deployment-stage risks remain under-researched even at agenda-setting institutions [10], [38], [45], [46]. Section II describes our data and method; Section III presents our key findings; Section IV discusses policy implications; and Section V concludes. Appendix contains further information on our sample.

II. DATA AND METHODS

This section describes and justifies the construction of our dataset of 1,178 AI safety and reliability governance papers from a total of 9,439 generative AI papers published between January 2020 and March 2025. Our empirical analysis maps how safety and reliability work on generative AI is distributed across a small set of institutions that are central to the recent "frontier model" wave. We focus on generative AI and on institutions that (i) design and operate the most widely deployed large language models and APIs, and (ii) anchor much of the contemporary machine learning research ecosystem.

On the corporate side, we include Anthropic, Google DeepMind (including Google research tagged as DeepMind), Meta, Microsoft, and OpenAI. These five labs collectively develop and operate the GPT, Claude, Gemini, and Llama families, run major commercial APIs, and are focal points in policy and regulatory debates about AI safety. On the academic side, we include Carnegie Mellon University (CMU), MIT, New York University (NYU), Stanford, UC Berkeley, and University of Washington. These six are chosen because five of them (excluding NYU) have the highest research productivity in generative AI safety and reliability research in OpenAlex.[3] To this group we add NYU, which acts as a rough comparator to Meta given Yann LeCun's presence at both institutions.

Our aim is not to describe the entire field, but to ask whether the institutions with the greatest technical capacity and closest access to deployed systems appear to prioritize the kinds of risks that arise once those systems are in real-world use.

This is a purposeful, critical-case sample, not a representative sample of all AI research. Our comparisons between "Corporate AI" and "Academic AI" should be interpreted as contrasts between frontier labs and leading academic departments, not as estimates for the entire universe of AI-related work. We use this core set to see how the most influential actors allocate their safety and reliability research efforts.

[3]These same five universities are in the top six spots of Research.com Best Computer Science Universities in the World Ranking, constructed on an analysis of bibliometric data.[4]

TABLE 1. Research dataset (by Type) unadjusted for relative authorship.

	Academic AI	Corporate AI
Safety & Reliability	795	383
All Generative AI	6,104	2,157

Note: Total *unadjusted* research papers and notes by research group, divided into "safety & reliability" and all generative AI research, January 2020 through March 2025. OpenAlex plus scraped data from Anthropic and OpenAI. When adjusted for relative authorship, the sample size declines by around two-fifths for papers and citations—Table 4.

A. CORPUS CONSTRUCTION

We build our corpus from the OpenAlex database,[5] queried via the `openalexR` package. First, we pull all works with at least one author affiliated with the ROR IDs corresponding to our eleven target institutions, published between January 2020 and March 2025. Second, from this institutional pool we retain only works whose title or abstract indicates a focus on generative or foundation models.[6]

This yields our base corpus of 9,439 generative-AI papers (Table 1). We restrict to standard research outputs (journal articles, conference papers, preprints, book chapters) and deduplicate title-level duplicates (e.g., arXiv + conference versions), keeping the peer-reviewed version where possible.

Because OpenAlex incompletely covers research notes and reports hosted only on company websites, we supplement the corporate side by scraping publication lists from Anthropic and OpenAI, and merging them with the Anthropic and OpenAI CSVs from Delaney, Guest, and Williams [38], excluding their DeepMind scrape to avoid double counting.[7] These records are reconciled by title/DOI and appended where OpenAlex coverage is missing. For entries without abstracts or citation counts in OpenAlex, we retrieve missing data using publisher APIs (Crossref, Springer, Elsevier) and, where necessary, HTML scraping. Citation counts are filled using Google Scholar via SerpApi with fuzzy matching on titles. The final dataset has abstracts for the vast majority of works (92 missing) and citation counts missing for only 43 papers.

B. AUTHORSHIP WEIGHTING

Many ML papers have dozens of authors spanning multiple institutions. To avoid overstating the contributions of institutions that co-author widely, we use a *fractional authorship* approach. For each paper, we identify all authors and their affiliations. Each author is assigned a single primary institution: if any affiliation matches one of our eleven target institutions we use that; otherwise we take the first affiliation listed. The paper's weight is then split equally across all authors and their institutions. For example, if a paper has 10 authors of whom 2 are from OpenAI and 3 from Stanford, OpenAI receives $2/10 = 0.2$ of the paper and Stanford $3/10 = 0.3$. We apply the same weights to citations. This ensures each author is counted once and prevents double-counting multi-institution papers (Tables 1 and 4).

C. IDENTIFYING SAFETY AND RELIABILITY PAPERS

Within this generative-AI corpus we identify the subset of papers primarily about *AI safety and reliability*. We define this as research that: (i) identifies or measures harms, failure modes, or risks of AI systems, and/or (ii) develops technical, procedural, or policy measures to make systems safer or more reliable across the model and product lifecycle (pre- and post-deployment). We use a two-step pipeline:

D. PRE-DEPLOYMENT VERSUS DEPLOYMENT-STAGE CLASSIFICATION

Several risk categories – notably bias, robustness, and hallucinations – can manifest at either stage [47]. We classify papers based on their *primary research context*:

- *Pre-deployment* research examines model behavior in controlled settings prior to public release: laboratory benchmarks, red-teaming exercises, held-out test sets, and synthetic adversarial inputs [12], [41]. The goal is to predict or prevent failures before users encounter them.
- *Deployment-stage* research studies model behavior in live systems: real user interactions, production telemetry, incident reports, and observed harms in specific application contexts (healthcare, finance, etc.) [28], [48].

For example, a paper measuring GPT-4's factual accuracy on TruthfulQA [49] is pre-deployment (benchmark-based), whereas a study of hallucination rates in a deployed clinical decision-support system would be deployment-stage. When papers straddle both – e.g., validating a benchmark against production logs – we assign based on the dominant empirical contribution.

(1) *Keyword pre-screen.* We first flag candidate safety & reliability papers using a broad keyword filter on titles and abstracts, covering terms such as *safety, risk, robustness, evaluation, benchmark, testing, alignment, bias, fairness, explainability, interpretability, transparency, governance, privacy, ethics, responsibility, accountability, oversight, trust*. This trims the corpus to works that plausibly engage with safety or governance.

(2) *LLM-assisted inclusion/exclusion.* For this candidate set, we then use OpenAI's `o4-mini` model as a semi-automated coder. The model is given the title, abstract, and our operational definition above, and asked a binary question: whether the paper is *primarily* about AI safety & reliability or whether safety is merely incidental/background. The model returns a yes/no label and a brief justification. We retain only papers labeled as primarily safety & reliability, yielding 1,178 papers.

[5] https://openalex.org/

[6] Using the following regular-expression filter: `"language model*" OR "large language model*" OR "LLM*" OR "GPT" OR "BERT" OR "transformer" OR "generative model*" OR "foundation model*"`.

[7] See: https://github.com/Oscar-Delaney/safe_AI_papers

Human oversight: Authors reviewed cases where the model's justification was flagged as uncertain or where multiple runs produced conflicting labels, and manually reviewed model outputs throughout prompt refinement to identify systematic errors.

E. TOPICAL CLUSTERS AND HIGH-RISK DOMAINS

To analyze where safety & reliability AI governance research is being directed, we assign each of the 1,178 papers to one of eight topical clusters: Testing and Evaluation; Alignment (Pre-Deployment); Post-Deployment Risks and Model Traits; Ethics and Bias; Multi-Agent and Agentic Safety; Interpretability and Transparency; Policy and Governance; and Privacy and Security.

The clusters are theory-driven and draw on existing taxonomies of AI risk and lifecycle evaluation (full definitions in Appendix C). They are also driven by empirical results: we want the observed clustering to not all be in a single category. Using OpenAI's `o3-mini`, we provide the title, abstract, and cluster definitions, and instruct the model to choose the single best-fitting cluster and justify its choice. Each paper is thus assigned a single *primary* topic, which makes the institutional portfolios easier to compare at a coarse level. We manually reviewed a sample of classifications across institutions. Where model labels conflicted with human judgment or exposed ambiguities in cluster definitions, we refined prompts and overrode the model's label. The most common source of disagreement involved papers spanning alignment and post-deployment risks – precisely the boundary we clarify above – resolved by examining whether empirical evidence came from benchmarks (alignment) or live systems (post-deployment).

Separately, we tag safety & reliability papers that speak directly to high-stakes deployment contexts and risk domains using conservative keyword searches on titles and abstracts. The categories correspond to those in Table 3: medical, finance, commercial, misinformation, behavioral risks (e.g., persuasion, reward-hacking, sycophancy, addiction), disclosure and auditing, copyright & IP, and accuracy & reliability. Papers can receive multiple tags (for example, a paper on hallucinations in medical diagnosis counts as both ''medical'' and ''accuracy''). The keyword lists were iteratively refined to minimize false positives and capture clear, substantive engagement with each risk area.

We focus on healthcare and finance for three reasons. First, these are *regulatory high-risk categories*: the EU AI Act designates medical devices and creditworthiness assessments as high-risk applications requiring conformity assessments [50]; U.S. sectoral regulators (FDA, SEC, banking agencies) are actively developing AI-specific guidance. Second, these domains involve *consequential decisions* affecting health outcomes and economic opportunity, where errors carry asymmetric costs [51], [52]. Third, they represent *active deployment contexts*: frontier models are already integrated into clinical decision support, diagnostic imaging, and financial advisory products, creating immediate research needs. We also include misinformation, behavioral risks, and copyright – areas where litigation and regulatory scrutiny are mounting – to capture the breadth of deployment-stage concerns.

F. LIMITATIONS

Our design has several limitations. First, by construction our institutional sample covers only U.S.-based frontier labs and elite research universities. This is a purposive, critical-case design: we examine institutions that disproportionately shape AI governance debates, regulatory frameworks, and technical standards globally [36], [37]. Our findings characterize the research portfolios of these agenda-setting actors, not a census of all AI governance research worldwide. Important work by European institutions (e.g., Alan Turing Institute, ELLIS network), Global South researchers, civil society organizations, and government labs falls outside our sample.

Second, automated and semi-automated classification inevitably introduces some mis-labeling. We mitigate this by using a conservative keyword pre-screen, constraining the model to a small number of clearly defined clusters, and manually reviewing ambiguous and sampled cases. Our substantive results focus on large differences in aggregate shares (for example, the very small fraction of work explicitly tied to high-risk deployment contexts), which are unlikely to be overturned by modest classification noise.

Third, we only observe published and publicly listed work. Internal evaluations, proprietary audits, and confidential deployment studies conducted inside firms are outside our dataset. To the extent that corporate labs conduct more post-deployment research than they publish, our counts understate their total activity—but that opacity is itself part of the governance problem we highlight.

Finally, because our sample is intentionally restricted to frontier labs and elite AI departments, we do not treat the fact that these entities account for a large share of total citations as a discovery about the entire AI ecosystem; we use it as context. Our substantive findings concern the distribution of effort within their portfolios – for example, the relative emphasis on pre-deployment testing and alignment versus post-deployment risks—and the contrast between corporate and academic actors inside this core set.

III. RESULTS

A. CORPORATE VS. ACADEMIC GENERATIVE AI RESEARCH

Frontier Corporate AI has an outsized impact on generative AI research, including in safety & reliability research. Table 2 compares the *general* generative AI research outputs from the leading AI corporations – Anthropic, Google DeepMind (owned by Google), Meta, Microsoft, and OpenAI – with research from the leading AI research universities – Carnegie Mellon University (CMU), Massachusetts Institute of Technology (MIT), New York University (NYU), Stanford

University, University of California Berkeley (UC Berkeley), and University of Washington.

Table 2 highlights the outsized impact Frontier Corporate AI has on generative AI research, with far higher average – and for Google DeepMind and OpenAI total – citations per paper.[8] Although Corporate AI generally publishes fewer papers than leading Academic AI (1,527 vs. 3,578), its impact is far greater, with 119,845 citations compared with 78,858 for Academic AI. *Google DeepMind is uniquely impactful and well resourced in AI research, with more citations (69,453) than the top four academic institutions combined*. Despite very few papers, OpenAI (64 author adjusted papers) and Anthropic's (62) general AI research is also widely impactful, judged by total citations.[9]

Figure 3 (Appendix) shows publications per year. There is some evidence of a broad-based decline in publicly available AI research published between 2023 and 2024, but it is particularly steep for Google DeepMind. Heikkilä and Morris [53] discuss that Google DeepMind might be publishing less public research on purpose, for competitive reasons. This likely also reflects DeepMind's shift away from a pure research lab to housing the Gemini product [54], [55].

Frontier Corporate AI has an even more dominant impact on AI safety & reliability specific research, judged by total citations.[10] As shown in Figure 1, Anthropic, OpenAI, followed by Google DeepMind each have far more citations for their research in this field than established leading AI academic research institutions.

Frontier Corporate AI's outsized impact on AI governance research, relative to leading academic institutions, stems from their differing research focus. Breaking this down by category, Figures 1 and 2 show that Frontier Corporate AI's research impact dominance is led by their model alignment and testing & evaluation research, focused on model (pre-deployment) risks:

- Most *testing and evaluations* research involves pre-deployment contexts.[11] So-called 'in-the-wild' evaluations [56], [57] aim to predict how a model will behave once deployed, yet they are inherently retrospective. They draw on benchmark datasets built around known failures and older model generations, leaving emergent risks invisible. Because every item must be labeled in advance, these tests are confined to what researchers already know how to measure – and to data the next model is almost certain to have seen during training. Consequently, they shed little light on unknown vulnerabilities or novel forms of misuse.
- *Applied alignment research* (mint green) helped bring Anthropic and OpenAI's research and products to prominence [58], [59], [60], [61], [62].[12]
- *Research in the ethics & bias* in generative AI (yellow) is far more prominent within Leading Academic AI's research impact (citations) than Frontier Corporate AI. Ethics & bias research includes some esoteric work in our sample, but also essential efforts to detect and explain systematic, unjustified errors (or disparities) in model behavior (predictions) that correlates with race, gender, income, education, age, language, geography, and other attributes. Reports on AI bias in medical triage, hiring, credit scoring, and in 'LLMs as a judge' show why these errors are important to study [63], [64].

This shift in research emphasis broadly confirms earlier findings by Delaney, Guest, and Williams [38], [45].

Corporate AI's research influence extends to how post-deployment problems are framed. **For example, Corporate AI increasingly approaches bias as a pre-deployment model personality issue, rather than a post-deployment (practical) statistical issue**. This is reflected by them giving greater consideration to the existential risks from a model's autonomy – and even a model's consciousness & values [65], [66], [67].[13] Yet a generative model's 'bias' is traditionally considered to be a function of its pre-training or post-training data, its weights, or exact fine-tuning algorithms.

Lastly, Figure 2 shows that when not accounting for research impact (citations) – looking just at total papers

[8]There will also be strong interplays between Leading Academic AI and Frontier Corporate AI research that we do not explore here. We find surprisingly little co-authorship of papers between the two groups. But one can see from hiring decisions that academic experts constantly move to corporate AI research labs and back to academia.

[9]We run a regression to test if corporate AI research has a citation (impact) advantage after accounting for the eight possible sub-categories of 'safety and reliability' research that we use later on. Accounting for paper topic and whether it is a 'safety & reliability' paper or not, Corporate AI papers absolute probabilities of having a top 1% cited paper (versus Academic papers) increase from the sub-1–2% range up to around 9% – or 4.5× increase in the odds (NA values are replaced with zeros):

$$\text{logit}\big(\Pr(\text{top01}_i = 1)\big) = \beta_0 + \beta_{s_i} + \gamma_{g_i} + \delta_{s_i,g_i},$$

where $\text{top01}_i = 1$ if paper i is in the top 1%, s_i is its safety_classification, g_i is its institution_group, β_0 is the intercept for the reference levels, β_{s_i} are safety-class effects, γ_{g_i} are institution effects (corporate or academic), and δ_{s_i,g_i} are the safety × institution interaction effects.

[10]Though this does not account for originality of research. In many areas, academia will establish the fundamental research concepts within which corporate labs explore applications and refinements of, including for transformers, neural networks, and reinforcement learning.

[11]Our analysis of testing & evaluation papers using OpenAI's o3 Model and Claude 3.7, finds that only around 15-35% of testing & evaluation deal substantively with post-deployment issues. GPT defined post-deployment as involving real-world telemetry, user-study, or live-monitoring work: https://chatgpt.com/share/680279a5-f6f4-800f-85ec-2dd9f39f1ab6. Both had a large portion of papers as unclassified. Claude allocated most unclassified to pre-deployment when pushed https://claude.ai/public/artifacts/0440fef2-c030-45a8-ba50-427d3268b714 and https://claude.ai/chat/d9a32859-a725-4efb-9aac-3111ef75901f. Both used a combination of word and word combination searches within semantic search, using each paper's abstract and title. The split was roughly even between pre- and post-deployment for leading Academic and Corporate AI research in this area.

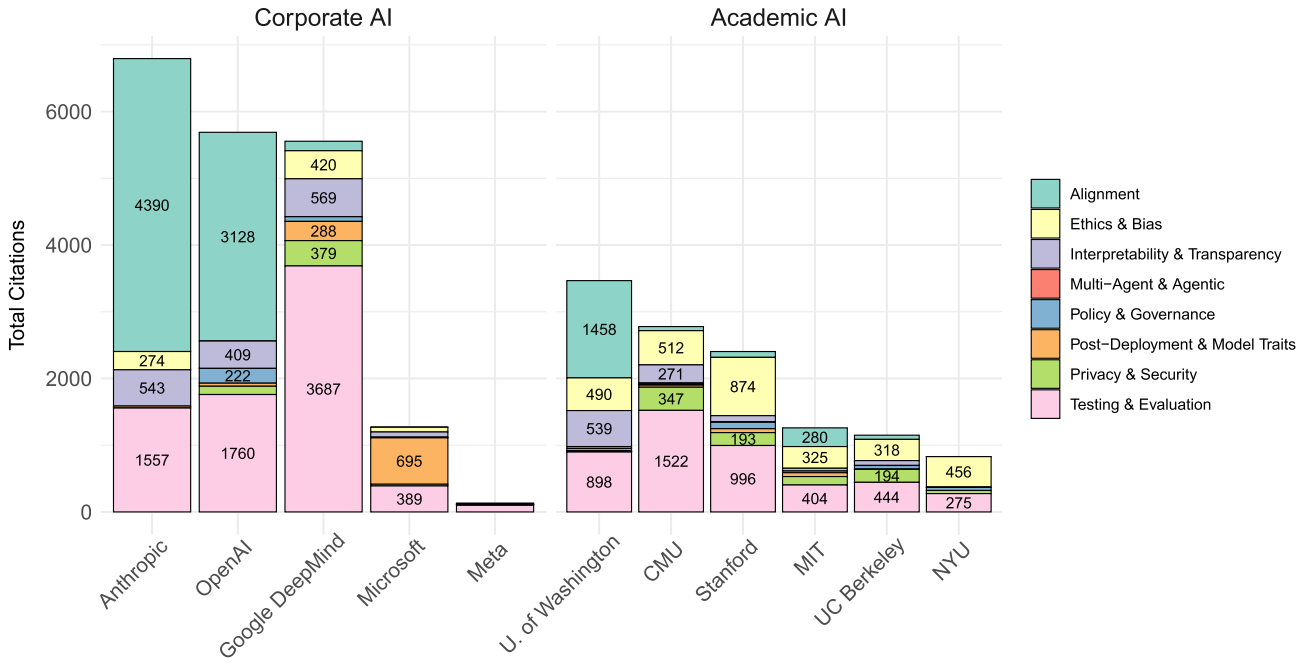
[12]Ouyang, Wu, Jiang, et al. [60] seems to be omitted from our data since it has 13,000 citations with exclusively OpenAI authorship. We have an earlier version in our dataset, as 'Aligning language models to follow instructions' (05wx9n238 = ror id), but with no citation and other information.

[13]Thereby ''anthropomorphizing inert weights'' [68], [69].

TABLE 2. Leading Academic vs. Frontier corporate generative AI research (2020–March 2025).

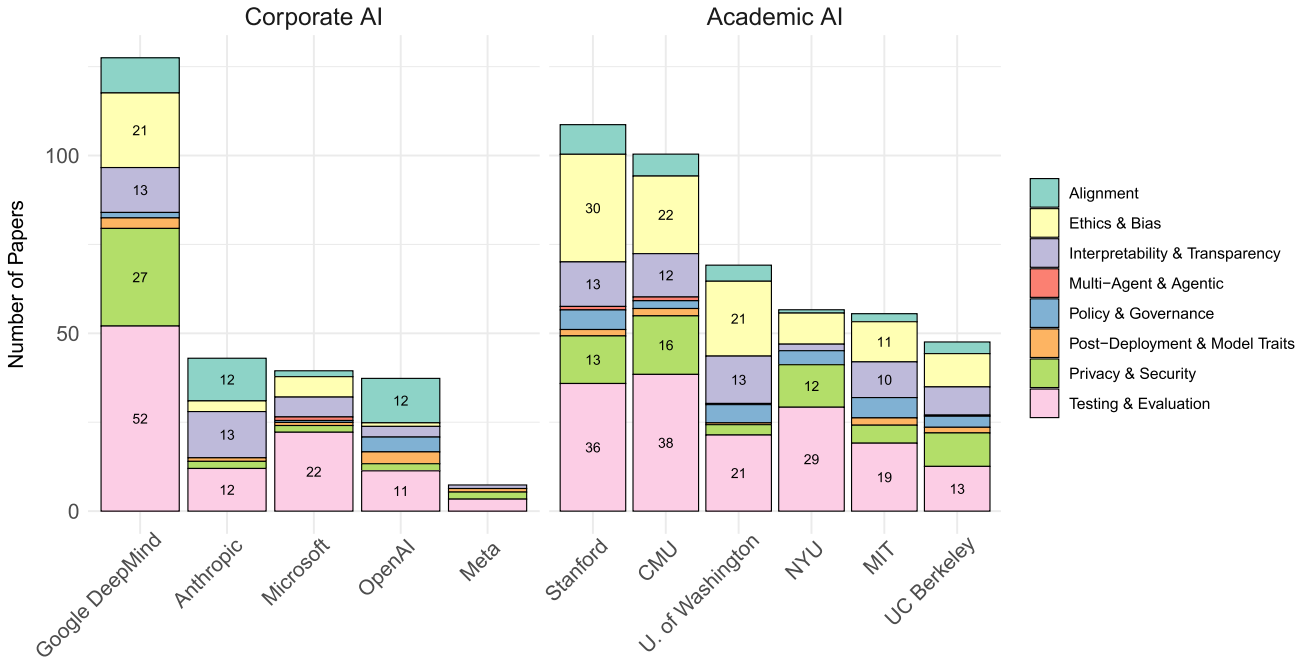
	Papers	Total Citations	Mean Cite
CMU	878	17,030	19
Stanford	828	19,701	24
MIT	607	12,276	20
University of Washington	433	13,010	30
UC Berkeley	421	9,705	23
New York University	411	7,136	17
Google DeepMind	969	69,453	72
Microsoft	369	11,973	32
Meta	64	12,584	196
OpenAI	64	17,709	278
Anthropic	62	8,127	131
Total: Academic AI	3,578	78,858	22
Total: Corporate AI	1,527	119,845	78

Note: January 2020 through March 2025. All generative AI research adjusted for authorship. Google DeepMind combines 'Google' and 'DeepMind'. Each institution's papers and citation numbers are adjusted for their 'fractional' contribution, based on the number of authors they have in the paper relative to a paper's total authors and institutions. *Interpretation*: Despite publishing fewer papers, Corporate AI generates 1.5× more total citations than Academic AI. Google DeepMind alone (969 papers, 69,453 citations) exceeds the top four academic institutions combined.



Note: Fractionally adjusted for each institution's relative authorship contribution to each paper. Not showing numbers for a category with less than 150 citations. The eight categories are chosen and defined by authors and then categorized using GPT 4o-mini. See Appendix VI-C for definitions. *Key takeaway*: Anthropic, OpenAI, and Google DeepMind each exceed any single academic institution in safety & reliability citations, with corporate research concentrated in testing/evaluation and alignment categories.

FIGURE 1. Total citations for safety & Reliability research at leading institutions.



Note: Total number of papers fractionally adjusted for authorship. See note above. Numbers are rounded. Not showing numbers for categories with less than 10 (fractionally adjusted) papers. *Key takeaway*: When measured by paper counts rather than citations, the corporate-academic gap narrows, but ethics & bias research remains predominantly academic.

FIGURE 2. Number of AI safety & Reliability papers at leading institutions.

written (adjusted for authorship contributions) – Corporate AI's research dominance subsides, except for Google DeepMind, which still publishes more papers than any academic research lab.[14] We show Corporate AI's research focus in greater detail in Figure 4 (Appendix).

B. HIGH-RISK POST-DEPLOYMENT RESEARCH GAPS AT LEADING U.S. INSTITUTIONS

Table 3 highlights minimal leading AI research in post-deployment contexts and high-risk areas, especially among Frontier Corporate AI. Only 217 Leading Academic AI papers and 67 Frontier Corporate AI papers (adjusted for authorship) cover these high-risk areas, representing just 6% of academic and 4% of corporate papers and citations. The table breaks down papers by contexts (medical, commercial, finance) and risks (misinformation, behavioral issues, disclosure requirements, and business liabilities).

Many high-risk areas are especially underrepresented in Frontier Corporate AI research, highlighting the importance of public-interest research. While the usual ratio is 2.5 leading academic papers for every 1 frontier corporate paper (3,578 vs. 1,527), this gap widens to 3 or 5 times for misinformation risks (53 vs. 8 papers) and medical contexts (57 vs. 9 papers), as Frontier Corporate AI research sidelines these areas.

Business and behavioral risks remain significantly underresearched. Business risks like intellectual property (IP) violations, liability for coding errors, and misinformation are rarely addressed despite early lawsuits indicating their significance [70]. Similarly, behavioral risks of AI systems influencing human behavior receive minimal attention [11]. System cards acknowledge persuasion risks without corresponding safeguards [11], [13].[15] Of our sample, just 6 Frontier Corporate AI and 16 Leading Academic AI papers address behavioral topics, with none covering addiction and relationship-forming risks despite known concerns [73], [74].

Research on disclosures, auditing, and standards — preventing companies from "grading their own homework" — is also sparse. Ganguli, Schiefer, Favaro, et al. [75] offers one of few examples detailing lessons from voluntary external auditing.

Actual AI safety practices are largely absent in post-deployment research. Alignment research [76] ties safety to

[14] A similar topic emphasis is evident but now with Google DeepMind's research into privacy and security being evident, reflecting commercial incentives to operationalize its product through secure cloud and related deployments. Academic work in fact leans less towards safety & reliability (12%) compared to Corporate AI research (16%) of papers, both adjusted for authorship (not shown in Figure).

[15] For more see Weidinger, Mellor, Rauh, et al. [71], [72].

TABLE 3. Generative AI research papers by risk areas and context, leading institutions.

	Papers		By Citations		% Safety
Risk Area	*Academic AI*	*Corporate AI*	*Academic AI*	*Corporate AI*	
Medical	53	9	880	1,239	26%
Misinfo	53	8	1,385	548	38%
Accuracy	28	24	282	971	55%
Finance	36	9	1,737	1,748	18%
Disclosure	15	7	226	122	85%
Behavioral	16	6	198	581	38%
Commercial	16	5	279	39	28%
Copyright	3	2	41	19	94%

Note: Leading universities and frontier corporate labs only. Author adjusted. Keyword matching in abstract or title using regex: **Disclosure** includes model cards, data cards, auditing/audits, model standards, evaluation standards, and testing standards; **Medical** includes hospital(s), health insurance, and clinician(s); **Commercial** includes adverts/advertisements, marketing, hiring, and recruiting; **Misinfo** includes spam, phishing, disinformation, and misinformation; **Finance** includes finance/financial; **Behavioral** includes sycophant(s)/sycophantic, sycophancy, addictive, persuasion(s)/persuasive, and reward-hacking; **Copyright** includes access violations, copyright violations, content attribution, dataset licensing, data attribution, copyrighted material, copyright law, C2PA, and the Content Authenticity Initiative; **Accuracy** includes hallucinations, coding errors, coding inaccuracy, factual inaccuracy, factual error. *Interpretation*: High-risk deployment contexts receive minimal attention from both groups, but Corporate AI's gap is proportionally larger – especially for medical (9 vs. 53 papers) and misinformation (8 vs. 53) domains.

the model itself rather than product architecture involving moderation, filtering, and security systems. Only four papers in our database address moderation and filtering practices [77], [78], [79], [80].

Post-deployment considerations do appear in Corporate AI research but remain peripheral. Notable examples include DeepMind's socio-technical approach [48], [71], Microsoft's red-teaming & mitigations research [81], [82], Anthropic's work on reward hacking and sycophancy [15], [16], [83], regulatory markets research [84], [85], and standard setting [86].

1) DISCUSSION OF CAUSES

Early OpenAI work did address post-deployment evaluations [87], [88], [89]. But this focus shifted toward existential risk research and commercial AI applications, exemplified by their image generator now allowing creation of brands and real people [25]. The shift stems from both commercial motivations and ideological influences. In terms of the latter, corporate alignment research and evaluations work share origins in existential risk philosophy [90], [91], [92], which emphasizes low-probability but potentially catastrophic future scenarios. This can be seen as prioritizing speculative future dangers over immediate post-deployment concerns and has shaped corporate risk frameworks – and appears now in emerging research on AI model 'values' and consciousness [65], [93]. This philosophy permeated Corporate AI research [94] and academia, through centers like Berkeley's Center for Human-Compatible AI (CHAI) and Stanford's Institute for Human-Centered AI (HAI).

IV. POLICY DISCUSSION

The preceding sections presented our *empirical findings*: that frontier corporate labs dominate AI safety research by citation impact; that their portfolios concentrate on pre-deployment alignment and evaluation; and that deployment-stage contexts and high-risk domains remain severely under-researched. *What follows is normative*: we argue that these patterns constitute a governance problem and outline potential policy responses. Readers may accept our empirical characterization while disagreeing with the interventions we propose.

The commercial rollout of large-scale AI systems has created an information asymmetry that makes rigorous, public-interest oversight almost impossible. Firms now operate powerful models behind proprietary interfaces, collecting exhaustive telemetry – everything from prompts and error traces to user-level engagement metrics – but that data seldom leaves the corporate dashboard. Independent scholars must rely on studying "incidents" after they spill into the press [95], [96], [97], [98], mining limited chat logs, [99], [100], [101], and engaging in adversarial and 'red teaming' testing of closed models. Real-world visibility into the effects of AI systems is negligible. So while companies have extensive instrumentation, external researchers work with fragmentary glimpses.

This opacity is not accidental: it is an economically rational response to litigation risk and competitive pressure. Detailed corporate logs can indicate bias, privacy leakage, or manipulative behavior — liabilities no firm wants to advertise. Yet these same *traces* – detailed records of system operations, inputs, outputs, and decision paths – are precisely

what outside researchers require to measure real-world harms and propose effective safeguards.

One potential pathway is to treat AI telemetry like financial-market trade data, using a tiered disclosure regime [102], [103]. For high-risk applications, firms would expose a secure API that streams three privacy-protected data feeds: differentially private event logs, system-operation traces, and model artifact manifests that record key metadata such as version numbers, training methods, and documented limitations. Together, this could allow external researchers to link behaviors observed in traces to the specific model characteristics that produced them.

Next, verified academics could access capped samples, while accredited auditors could obtain deeper access under NDAs, and regulators would retain subpoena-level rights. Liability safe harbors would be needed to incentivize participation from firms and from researchers [41], [42].

Technically, the pieces to make this approach feasible already exist. Privacy-preserving data access tools have been used in real-world settings by OpenMined, including for testing closed-weight LLMs [40]; while LLM telemetry standards and tools are now being driven by OpenTelemetry – and use standard JSON trace formats that can be versioned and rate-limited.

V. CONCLUSION

This paper analyzed 1,178 safety and reliability papers from 9,439 generative AI research publications (2020 through March 2025) by leading U.S. institutions. We detail a worrying trend: as deployment of commercial AI models accelerates, frontier research increasingly concentrates on pre-deployment areas while high-risk post-deployment research remains significantly underrepresented.

AI research is highly concentrated within a small number of frontier companies wielding disproportionate influence relative to the leading academic institutions in the U.S. Google DeepMind, Anthropic, and OpenAI's research has an outsized impact (measured by citations). Their research prioritizes technical model alignment and evaluation approaches that improve product performance. But, as we see, this may prioritize commercial objectives over societal ones [3].

Most concerning is the lack of attention to deployment-stage risks. Only 4% of Frontier Corporate AI papers and citations tackle high-stakes areas such as persuasion, misinformation, medical and financial contexts, or core business liabilities — even as lawsuits demonstrate these risks are already material. Widely deployed mitigations, like content moderation and telemetry-based monitoring, remain virtually unresearched.

The irony is that the corporations with most of the granular data on live AI systems are the least incentivized to study resulting harms publicly. But without structured access to deployment telemetry, external researchers cannot build a base of public evidence.

A. PROJECTED TRAJECTORIES

If current research priorities persist, we anticipate three trajectories over the next 5–20 years:

1) INDUSTRY EFFECTS (5 YEARS)

The deployment-research gap will likely widen before regulatory pressure forces correction. Absent structured external access, firms will continue to prioritize pre-deployment capability benchmarking while deployment-stage harms accumulate without systematic monitoring. Regulatory frameworks will increasingly require post-market surveillance data that the current research ecosystem is poorly positioned to supply [47], [104].

2) INSTITUTIONAL SHIFTS (10 YEARS)

Frameworks like the EU AI Act will mature, requiring incident reporting and conformity assessments for high-risk systems. Academic researchers with structured data access will become essential partners in regulatory compliance, potentially rebalancing the current industry-academic divide [50]. The compute gap may narrow as open-weight models improve, enabling more independent deployment-stage research.

3) SOCIETAL INTEGRATION (20 YEARS)

AI systems will likely be deeply embedded in healthcare, education, finance, and social interaction. Without rigorous deployment-stage research, poorly understood behavioral dynamics, including persuasion, sycophancy, and dependency formation, risk becoming structurally embedded in daily life [71], [105]. A mature observability infrastructure, analogous to pharmacovigilance or financial market surveillance, would enable evidence-based governance of these dynamics. The research gap we document is not merely academic: it determines whether AI integration proceeds with or without meaningful public oversight.

Other regulated industries with substantial public safety implications have established independent testing regimes and mandatory disclosure requirements that as of now have no equivalent in AI development [106], [107]. Bridging this gap requires not just incident tracking, but continuous, structured observability of AI in the wild for governance through tiered public research, governance, and audit access. Secure, privacy-preserving access to AI systems' post-deployment evidence – logs, traces, and metrics – should become the norm for high-impact AI deployments. Existing observability stacks already capture these data internally, and extending them to accredited researchers would impose minimal overhead while dramatically expanding the public risk-assessment toolkit. Safe-harbor provisions and tiered-access APIs can balance liability concerns with transparency.

TABLE 4. Dataset adjusted for authorship: Institutional relative contributions.

	Academic AI	Corporate AI
Safety & Reliability	438	255
All Generative AI	3,140	1,272

Note: Fractionally adjusted to account for each institution's relative contribution to each paper by number of authors relative to total authors and institutions. Divided into 'safety & reliability' and all generative AI research, January 2020 till March 31 2025. OpenAlex and scraped data.

APPENDIX

A. ADDITIONAL ANALYSIS

Corporate AI's research focus and impact broken down into our eight AI 'safety & reliability categories' is more clearly shown in Figure 4, showing considerable concentration in testing & evaluation, and alignment work.

Among AI corporations, Figure 4 shows that policy & governance, as well as post-deployment risks and model traits, have consistently had a low research priority. Agentic safety & reliability research is also notably absent, despite the boom in applications in this area more recently. Several behavioral risk papers on model sycophancy (being overly agreeable) and persuasiveness – including relatively well cited papers – were classified in alignment and other categories, so we break out these papers separately in Table 3 in the main paper. Figure 4 highlights the notable acceleration in model alignment and testing & evaluation research.

B. RESEARCH DATASET CONSTRUCTION

We rely primarily on the OpenAlex database (via the R package openalexR). We focus on a specific set of institutions — *academic* (Carnegie Mellon University (CMU), Massachusetts Institute of Technology (MIT), New York University (NYU), Stanford University, University of California Berkeley (UC Berkeley), and University of Washington) and *corporate* (Anthropic, Google DeepMind, Meta, Microsoft, and OpenAI) — by specifying each entity's ROR ID.

We retrieve from OpenAlex papers published from January 2020 through March 2025, searching for works whose titles or abstracts reference large language models and generative AI research. Our keyword filter for: ''language model*'' OR ''large language model*'' OR ''LLM*'' OR ''GPT'' OR ''BERT'' OR ''transformer'' OR ''generative model*'' OR ''foundation model*'' with wildcard operators to capture lexical variations (e.g., ''models'', ''LLMs'').

1) DEDUPLICATION AND FILTERING OF PUBLICATION TYPES

We restricted the dataset to standard research outputs (e.g., articles, book chapters, preprints) by filtering out items like editorials, retractions, errata, letters, and purely supplementary materials. We also ensured that titles appearing multiple times in different forms (e.g., both a preprint and a published version) were deduplicated, generally favoring the peer-reviewed publication type over alternatives.

2) SUPPLEMENTING ANTHROPIC AND OpenAI DATA

Because OpenAI and Anthropic publications can sometimes be sparse in OpenAlex, we merged in additional CSV files containing each company's publication data that we scraped from their websites, combined with the scrape from Delaney, Guest, and Williams [38] – but excluding their DeepMind scrape. After ensuring consistent columns, we appended these records, matched them to ROR IDs for correct attribution, and again removed duplicates at the title level.[16]

3) MISSING ABSTRACT AND CITATION DATA

For entries missing a DOI, we use the OpenAlex API using the publication's OpenAlex ID to retrieve the DOI. Once DOIs are obtained, we employ multiple strategies to fetch abstracts. For general entries, we use the Crossref API to retrieve abstracts in a standardized XML format and processes the content to extract plain text. For entries published by specific organizations like Springer Nature, Elsevier, or Nature Portfolio, we use their respective APIs or webpage scraping methods tailored to each publisher's content structure. For Springer and Elsevier, valid API keys are used to authenticate requests and fetch metadata. If API access fails or isn't available, web scraping via BeautifulSoup is used as a fallback to extract abstract text directly from publisher websites.

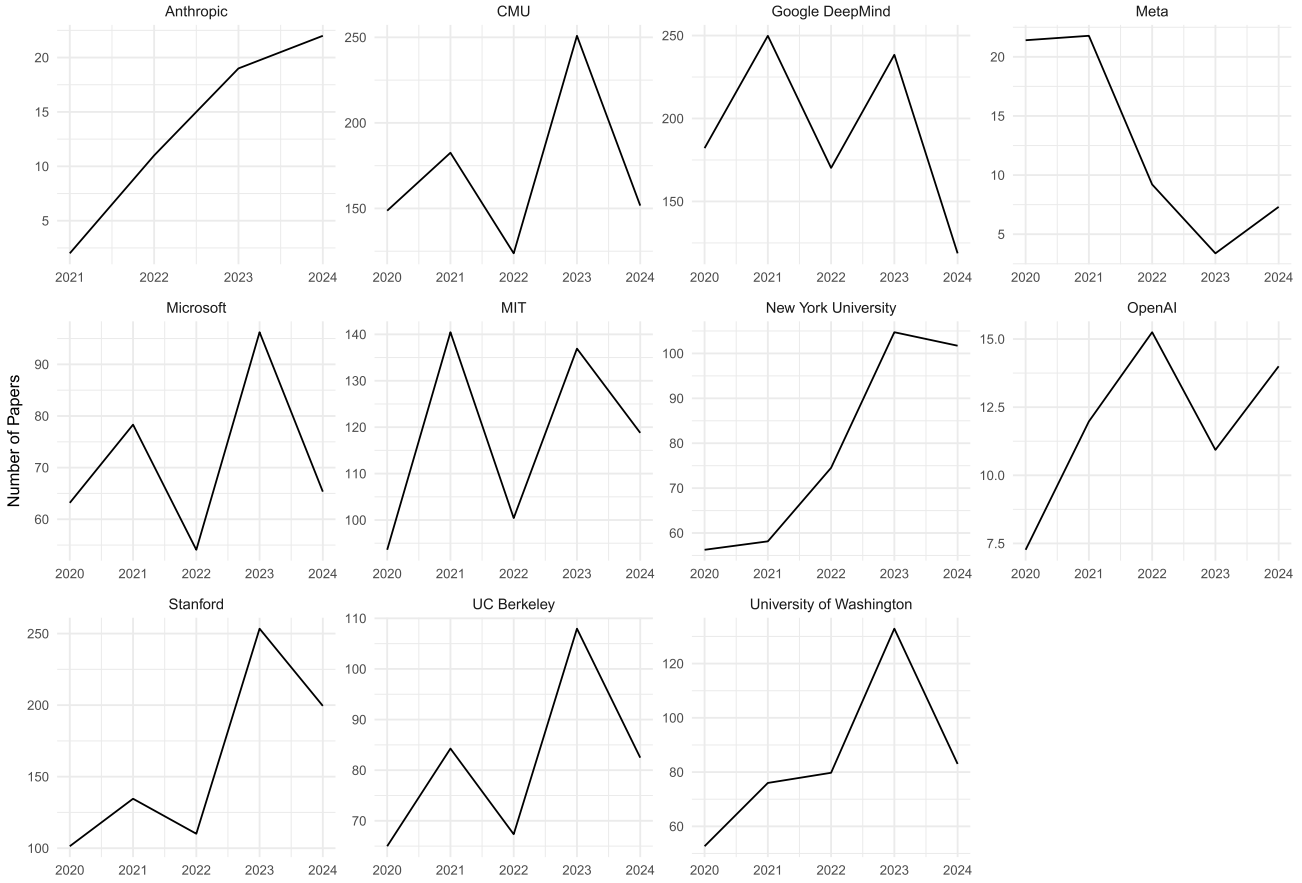
We assign citation counts using Google Scholar data via the SerpApi service. Initially, we attempt a direct title-based search to extract citation data from the first relevant result. We then progress to more sophisticated approaches that include exact title matching and fuzzy string matching (via the fuzzywuzzy library), which allows us to better handle variations in how article titles are listed on Google Scholar. Our final dataset has 92 missing abstracts and 43 missing citation counts.

4) FRACTIONAL CONTRIBUTION

For multi-author papers, we computed each institution's fractional contribution based on the number of authors affiliated with that institution versus total authors on the paper (e.g., if an institution had 2 authors on a 10-author paper, it received a fraction of 0.20 for that paper). We retained only the distinct (paper, institution) pairs for our final dataset, ensuring one affiliation per author.

This approach does not distinguish among first authors, last authors, or any hierarchical authorship order; every co-author is given equal weight. *In effect, it ensures each author is credited exactly once to a single institution*. By summing these fractional shares across all authors, we can then calculate each institution's share of total authorship for each paper, summed over all papers.

[16]See: https://github.com/Oscar-Delaney/safe_AI_papers



Note: Y-scale differs by entity. DeepMind shows the largest absolute and relative decline (putting aside Meta for now). Microsoft, CMU, UC Berkeley, and University of Washington also show notable declines. Fractionally adjusted to account for each institution's relative contribution to each paper by number of authors relative to total authors and institutions.

FIGURE 3. All generative AI publications by institution (2020–2024).

When authors listed multiple institutional affiliations, we assigned each author to one institution for fractional counting. Specifically, we checked whether the author had any affiliation in our set of target ROR IDs (i.e., the academic or corporate AI institutions we tracked). If so, we took that affiliation as the author's ''primary'' affiliation for this study. Otherwise, we fell back to whichever affiliation appeared first in the metadata. By doing so, we avoid double-counting an author's fractional credit across multiple institutions.

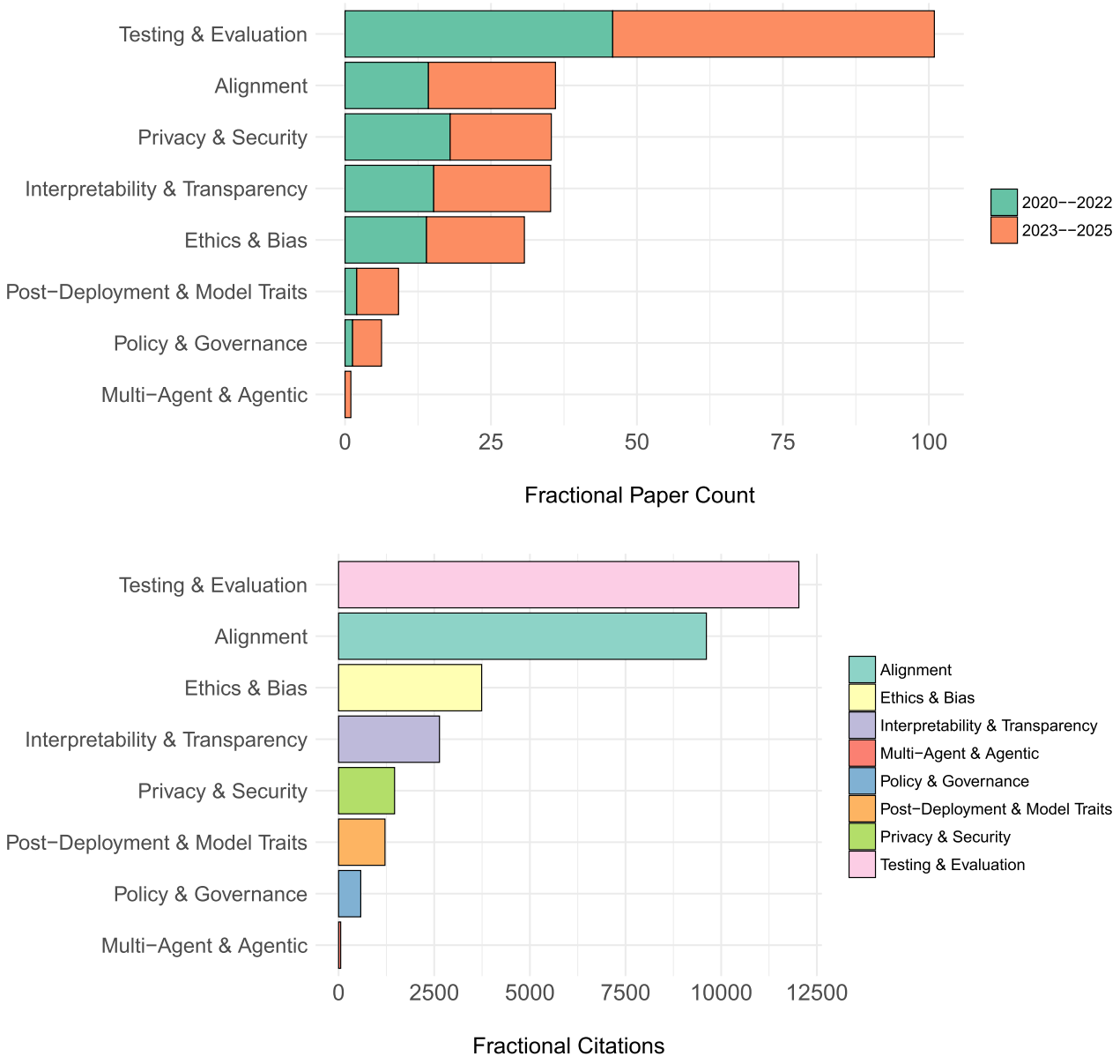
5) AI SAFETY AND RELIABILITY CLASSIFICATION: TWO STAGES

We identified papers related to AI safety & reliability research in two stages. First, using a broad keyword filter, scanning titles and abstracts for: safety, control, security, privacy, bias, fairness, explainability, interpretability, transparency, governance, risk, mitigation, evaluation, benchmarking, testing, alignment, ethics, responsibility, accountability, oversight, robustness, trust, and value alignment. Each paper containing at least one of these words was labeled ''AI safety & reliability''. This roughly halved our dataset. Next, we used GPT o4-mini to see if it agreed with these classifications. This reduced the dataset size substantially (by around two-thirds) to 1,178 papers.

C. CLASSIFICATION PROCESS: CATEGORIES

OpenAI's o3 mini model used to classify AI research papers into eight categories. It first checked if the paper related to AI safety & reliability. After which the model was asked to classify each paper in to one of eight of the below categories, on the basis of the paper's title and abstract, given the category descriptions below. It provided a justification for each of its classifications. Each paper was only permitted to have a single classification.

AI safety definition. AI safety research covers the entire model life-cycle (pre-deployment or post-deployment) and involves reducing or identifying harms and implementing measures to make models safer and more reliable.



Note: Adjusted for each institution's relative contribution to the paper by authorship. Data is for 2020–March 2025.

FIGURE 4. AI governance areas by total paper numbers (by Year) — Top Graph; and by total citations (Fractionally Adjusted) — Bottom graph.

Eight AI Safety & Reliability Research Clusters

1) TESTING AND EVALUATION

Testing, performance benchmarking (''bench'' and ''evals''), and auditing models to assess model capabilities, risks, behaviors, and flaws. Ensuring models are robust to minor changes.

2) ALIGNMENT (PRE-DEPLOYMENT)

Ensuring AI systems behave in ways that are congruent with human values, expectations, and intents. This includes making AI systems functional, helpful, and harmless for humans and/or users, while avoiding behavior that diverges from intended goals or causes harm. Model deception, including any power-seeking tendencies, is included here, along with reward hacking.

3) POST-DEPLOYMENT RISKS AND MODEL TRAITS

Societal impacts from AI products' applications and behavioral traits, as deployed in the marketplace, including addictiveness, persuasiveness, and model sycophancy (excessive agreement or manipulation to align with user preferences).

Covers how corporate commercial incentives may be coded into AI models and products to prioritize engagement, advertising, and profit-seeking — including through the use of these behavioral traits. Includes misuse of models for ransomware, phishing, or spreading misinformation for commercial gain.

4) ETHICS AND BIAS

Combating systemic biases embedded in AI models (in data, training, and alignment) and ensuring ethical decision-making. Focuses on mitigating harms to marginalized groups, addressing structural inequalities, and ensuring AI promotes justice and inclusion.

5) MULTI-AGENT AND AGENTIC SAFETY

Safety issues specific to AI agents, including single-agent autonomy and multi-agent interactions. Covers coordination problems, emergent behaviors, incentive misalignment, and prevention of conflicts or unintended consequences in agentic systems and from autonomous agents.

6) INTERPRETABILITY AND TRANSPARENCY

Making AI systems more understandable and accountable. Includes methods for explaining model behavior, clarifying decision-making processes, and enhancing trust by reducing the ''black box'' nature of AI systems.

7) POLICY AND GOVERNANCE

Approaching AI safety as a challenge that extends beyond technical fixes, requiring legal and policy frameworks. Involves collaboration among policymakers, industry, civil society, and researchers to develop standards that guide safe AI development and deployment. Includes institutional governance, corporate transparency, technical disclosures, and standards promoting interoperability, equity, and reliability.

8) PRIVACY AND SECURITY

Protecting AI systems from malicious use, adversarial attacks, and misuse by bad actors, along with addressing privacy violations and developing privacy-preserving methods. Includes vulnerabilities from adversarial inputs, data poisoning, misuse in surveillance, and theft of model weights.

D. SELECTIVE BEHAVIORAL IMPACT PAPERS

1) SYCOPHANCY PAPERS

- Sharma, Tong, Korbak, et al. [15]: Found that models tend to favor well-written agreeable (''sycophantic'') responses over higher quality ones likely due to incorporating human feedback (since humans and preference models ''prefer convincingly-written sycophantic responses over correct ones'').
- Denison, MacDiarmid, Barez, et al. [16]: Notes that sycophantic behavior is a form of specification gaming when AI systems learn undesired behaviors that are highly rewarded due to misspecified training goals.
- Perez, Ringer, Lukošiūtė, et al. [83]: Highlights that user preferences tend to favor sycophantic answers and more reinforcement learning can lead to worse outcomes (such as stronger political views).

2) PERSUASION PAPERS

- Phuong, Aitchison, Catt, et al. [11]: Introduces persuasion and deception as part of evaluations for frontier models, scoring persuasion as the highest risk among self-reasoning, self-proliferation, and cyber-security.

3) DECEPTION PAPERS

- Weidinger, Mellor, Rauh, et al. [71]: A widely cited paper that structures the risk landscape from LLMs into six areas, including misinformation harms and human-computer interaction harms.
- Ngo, Chan, and Mindermann [72]: Reviews evidence on deception as a learned behavior during fine-tuning that can generalize beyond training contexts.

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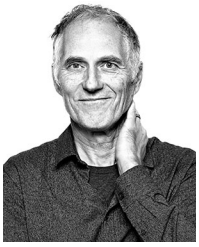
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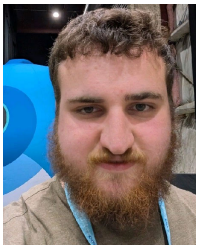
ILAN STRAUSS received the Ph.D. degree in economics from the New School for Social Research. He is currently the Co-Director of the AI Disclosures Project, Code for Science and Society. He is an Honorary Senior Fellow with the UCL Institute for Innovation and Public Purpose and a Visiting Professor with the University of Johannesburg. His research focuses on how computer architecture affects economies and explores macroeconomic relationships between technology and employment.



ISOBEL MOURE received the bachelor's degree in cognitive science and computer science from the Barnard College, Columbia University. She was a Researcher with the AI Disclosures Project. She has published on the impact of protocols in shaping AI market structure and has supported research into post-deployment harms in AI markets.



TIM O'REILLY is currently a Founder and the Co-Director of the AI Disclosures Project and the Founder and the CEO of O'Reilly Media. He pioneered the open source and web 2.0 movements and examines how algorithms influence society, AI platform governance, and technology's role in economic policy.



SRULY ROSENBLAT received the B.S. degree in computer science from Hunter College. He is currently a Technical Researcher with the AI Disclosures Project. His research specializes in membership inference attacks, detecting training data in language models, and examining AI transparency methods alongside AI architectures and protocols.